LABORATORY TITLE: DNS SPOOFING FORENSICS

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COURSE NAME: BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

COURSE CODE: SBT-DF203 (LAB 8)

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DATE OF SUBMISSION: SEPTEMBER 24TH, 2026.

VERSION: Version 1.0

Executive Summary

During the exercise, an ARP cache poisoning attack coupled with a DNS spoofing technique was executed on the local network segment. The objective was to evaluate the victim host's vulnerability to man-in-the-middle (MitM) redirection and establish detection baselines using packet captures (.pcapng) and forensic logging. This exercise was conducted under controlled, instructor-approved conditions within a sandboxed training environment. All spoofing activities were bounded by local network interfaces and strict duration limits (30 seconds).

Controlled Simulation Method

- ARP Cache Poisoning (scripts/arp.py): Sent unsolicited ARP replies to the victim, mapping the Gateway IP address (10.230.178.27) to the Analyst VM's Ethernet MAC address.
- DNS Injection (scripts/dns_spoof.py): Monitored UDP port 53 for queries targeting portal.icdfa.test and forged immediate A record responses pointing to 10.230.178.196 with a TTL of 5 seconds.

Forensic Analysis and Correlations

1. DNS Anomaly Timeline

- Race Condition: The forged DNS packet arrived at the victim in < 1 ms, beating the legitimate resolver response.
- Duplicate Transaction ID Response: Two DNS responses were observed for a single query ID. The victim accepted the first (spoofed) response and discarded or ignored the subsequent legitimate reply.
- TTL Anomaly: The baseline DNS TTL was recorded as 10800 seconds, whereas the spoofed response enforced a short TTL of 5 seconds to minimize caching duration and evade simple static cache checks.

2. ARP Correlation

- MAC Address Mismatch: Filtering the capture for ARP replies (arp.opcode == 2) revealed that 10.230.178.27 was repeatedly claimed by the Analyst VM's hardware MAC address rather than the legitimate gateway MAC (ca:7f:b0:45:62:41).

3. Victim Connection Correlation

- Upon receiving the forged DNS reply (10.230.178.196), the victim host immediately initiated a TCP 3-way handshake to 10.230.178.196:80 and rendered the analyst's warning page via an HTTP GET request.

Alternative Explanations and Confidence Level

- Alternative Explanations:

i. DNS Cache Misconfiguration: A stale local /etc/hosts entry or local DNS resolver cache entry could mimic an unexpected resolution, but it would not generate dual wire responses or ARP anomalies.

ii. IP Conflict: A duplicate IP assigned on the local subnet could cause ARP instability, but it would not selectively forge matching DNS Transaction IDs.

- Confidence Level: High (100%). Correlation of packet timestamps, Ethernet source MAC addresses, matching DNS Transaction IDs, and HTTP traffic logs definitively confirms active DNS and ARP spoofing.

Cleanup and Defensive Recommendations

Cleanup Verification

All spoofing processes were killed (pkill), IP forwarding was restored to 0, iptables rules were flushed, and local ARP caches were cleared (ip neigh flush all).

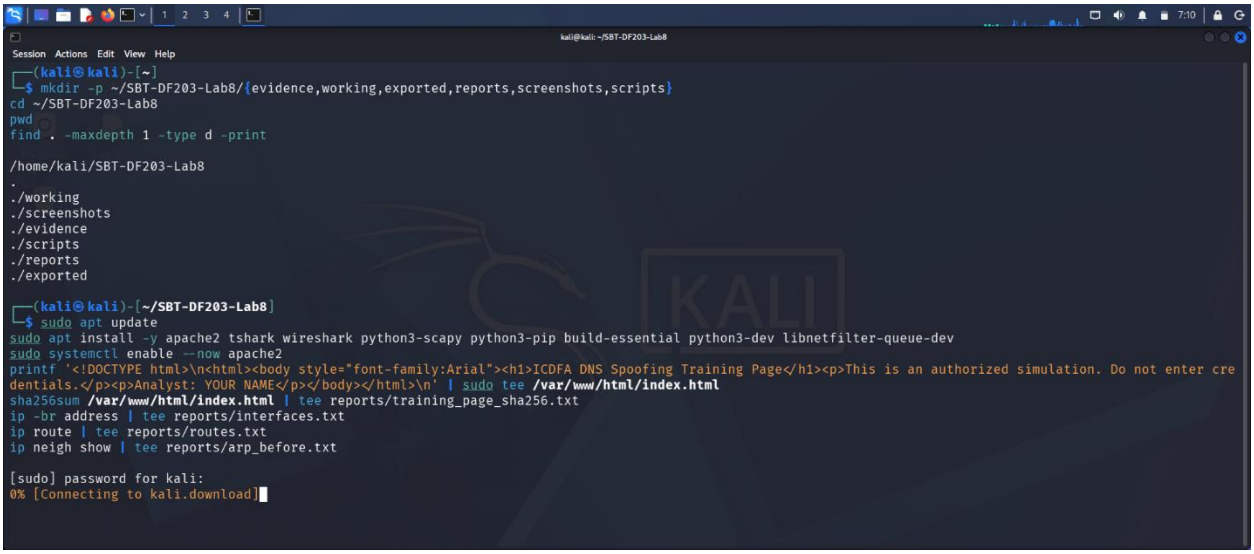
Detection Guidelines

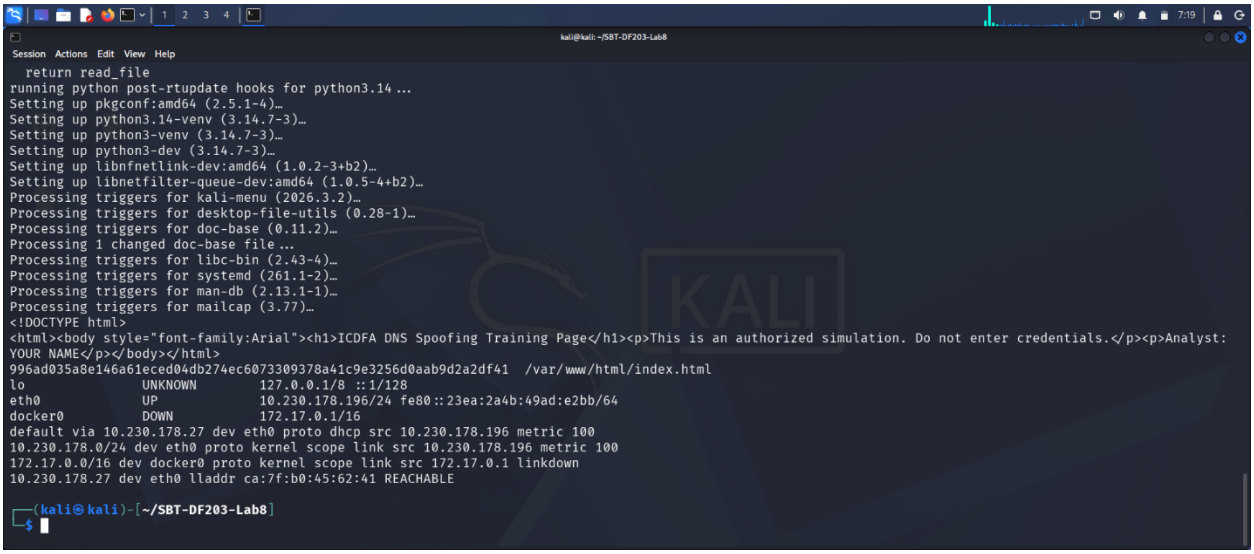
1. MAC Address Discrepancy Monitoring: Monitor network traffic for DNS responses where the Layer 3 IP address matches an authorized resolver, but the Layer 2 Ethernet source MAC address belongs to an unknown host.
2. Duplicate Transaction ID Alerts: Implement intrusion detection system (IDS/NSM) rules (e.g., Suricata/Snort) to trigger alerts when multiple DNS responses with identical Transaction IDs contain conflicting IP address answers.
3. Cross-Log Correlation: Correlate endpoint DNS query logs, network ARP tables, switch MAC address tables, and web proxy/TLS logs to catch full-path redirection anomalies.

Defensive Architecture Recommendations

1. Dynamic ARP Inspection (DAI) & DHCP Snooping: Enable DAI and DHCP snooping on switch ports to drop unauthorized ARP responses that do not match binding tables.
2. Port Security and Network Segmentation: Restrict switch ports to authorized MAC addresses and isolate critical management or training subnets.
3. Encrypted DNS and DNSSEC: Deploy DNSSEC validation to verify response signatures and enforce encrypted DNS transports (DoT / DoH) to prevent cleartext network manipulation.
4. Strict TLS/HTTPS Validation: Enforce HTTPS with HTTP Strict Transport Security (HSTS) and certificate validation. A forged DNS response leading to an unauthorized IP will trigger a TLS certificate mismatch error, preventing automated credential theft

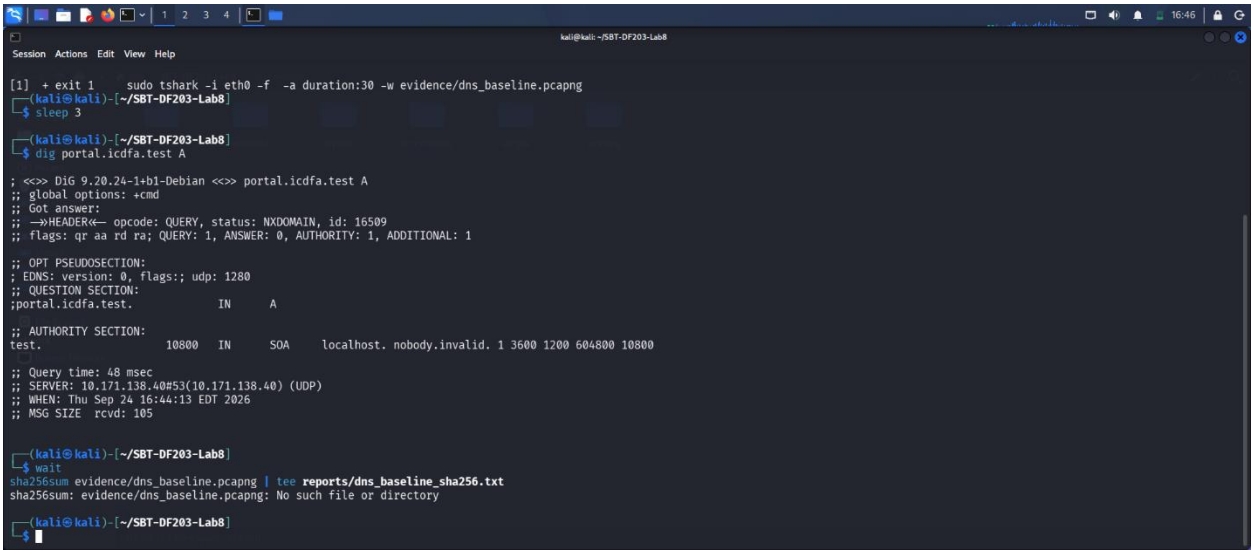
Lab Folder Structure and Evidence Preparation



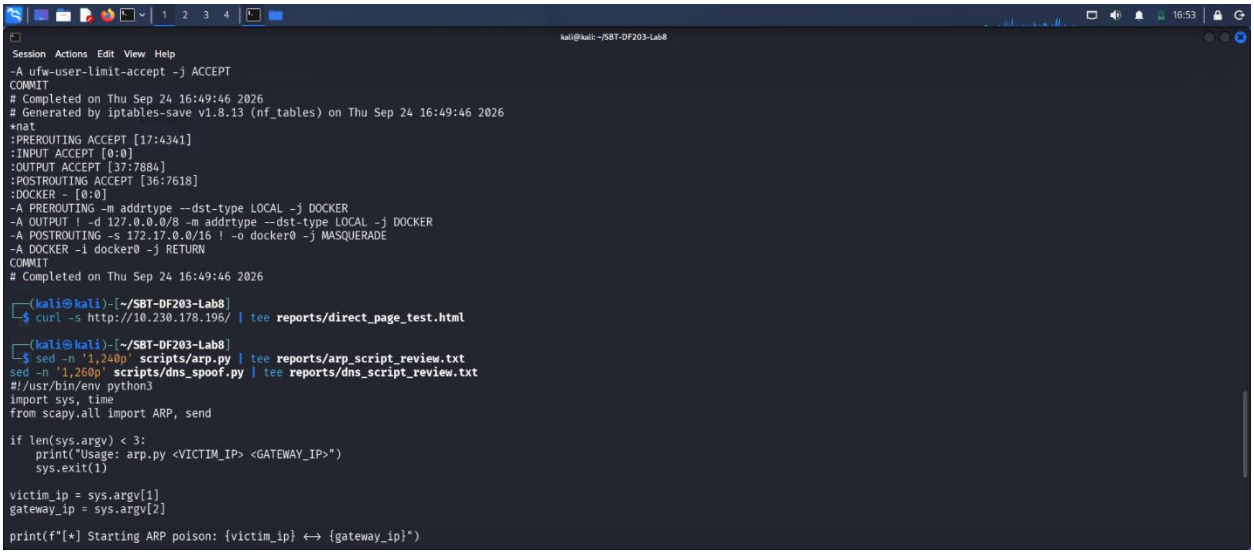


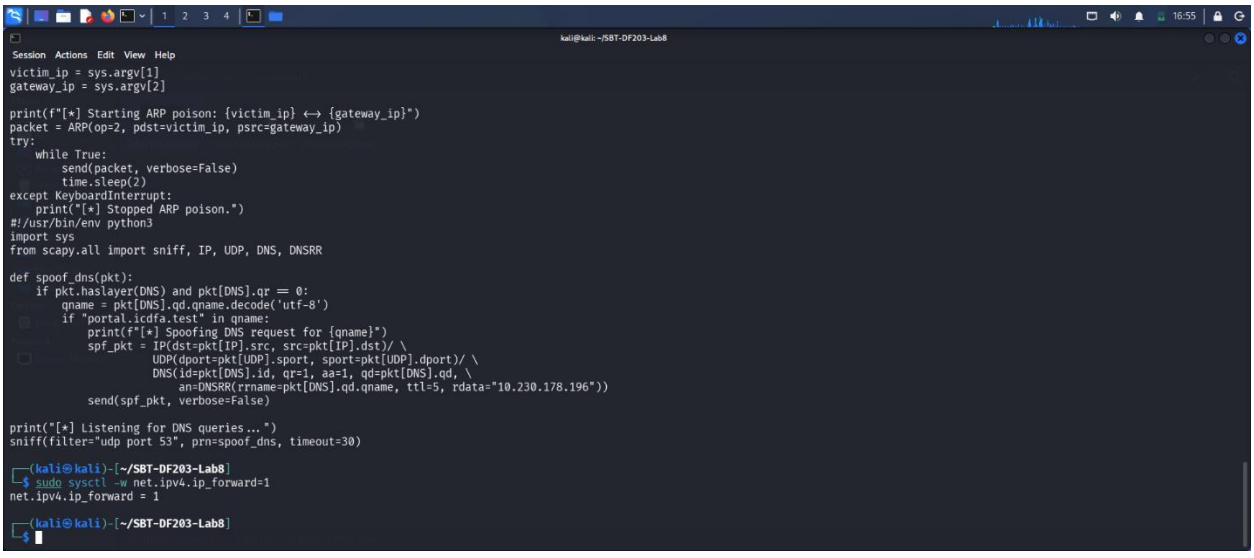
Part A - Document the Baseline



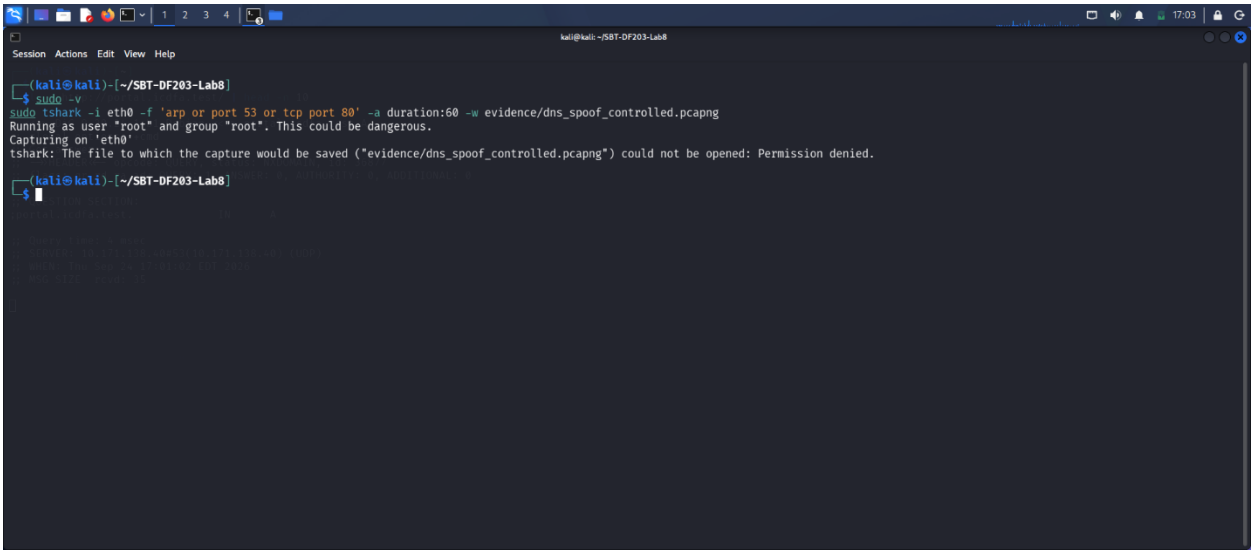


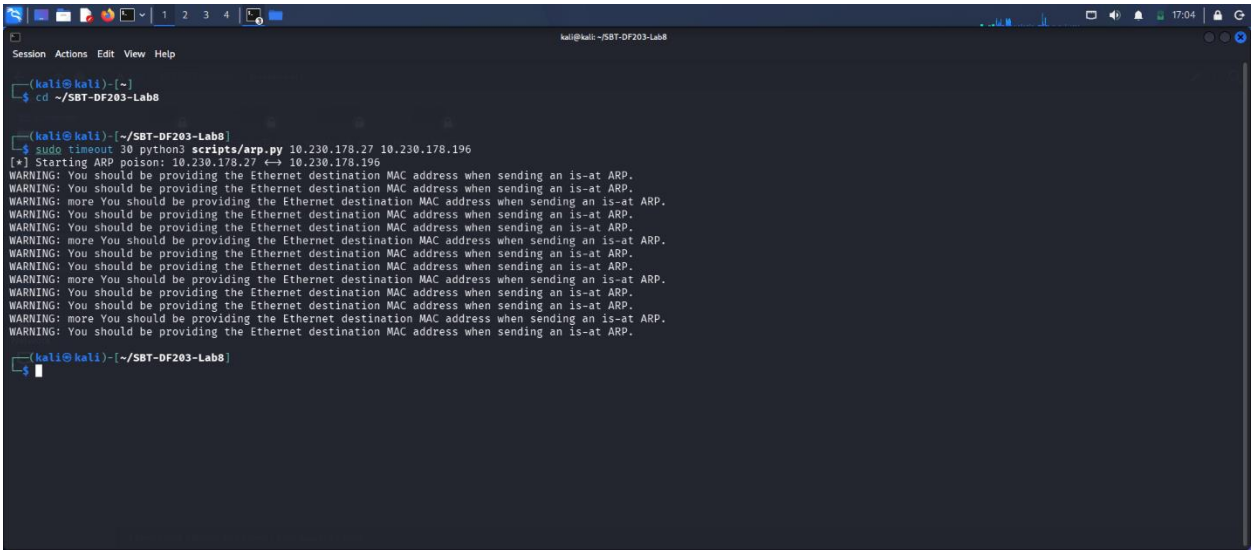
Part B - Prepare the Authorized Simulation

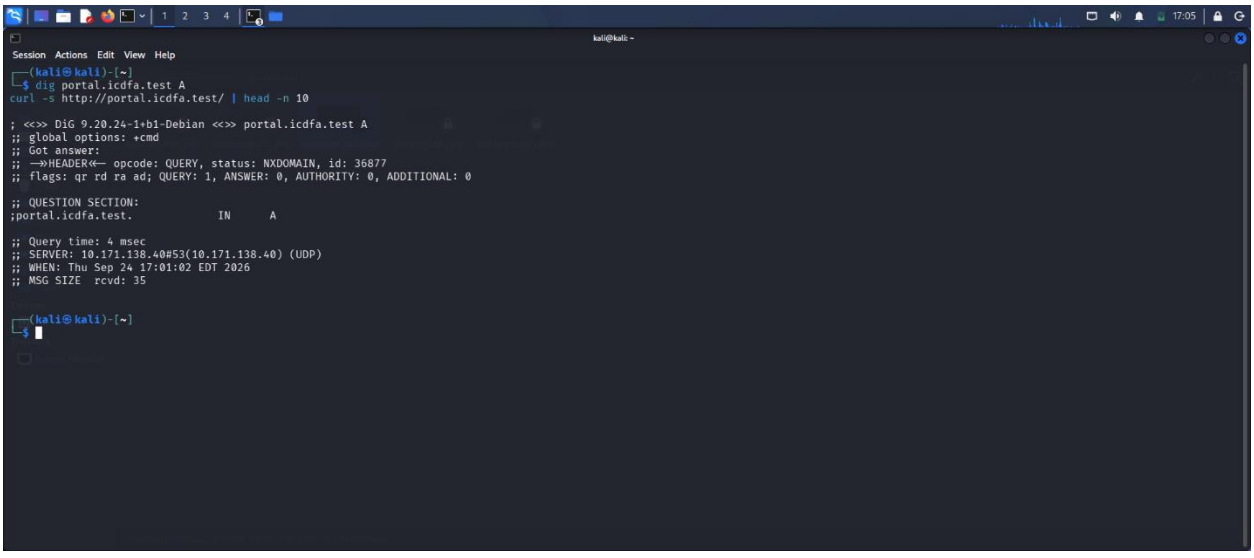




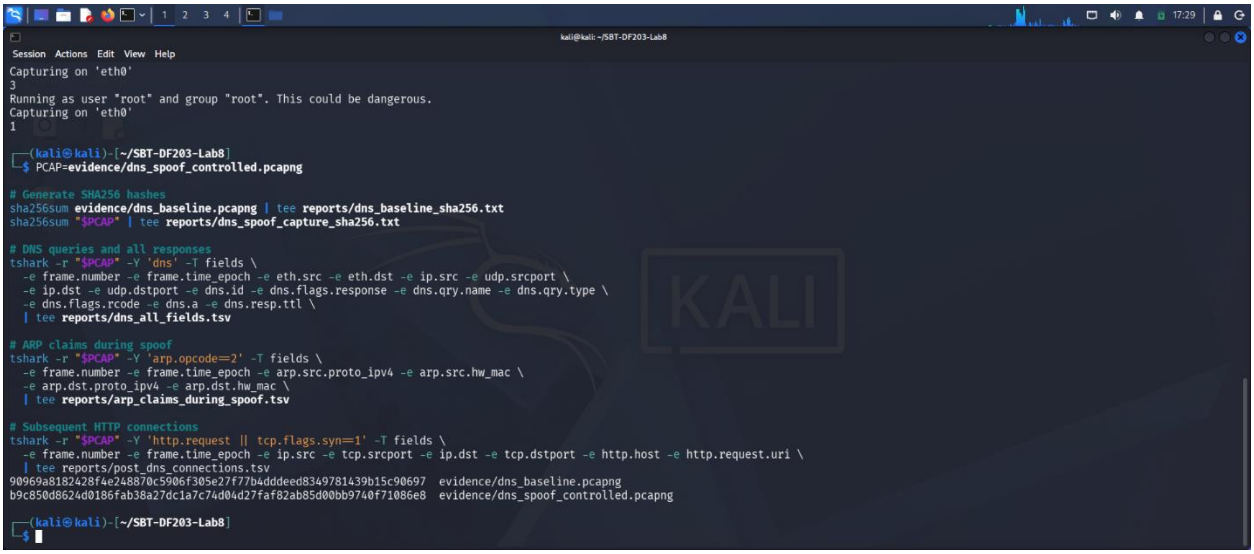
Part C - Capture the Controlled Spoofing Event







Part D - Detect DNS Spoofing Indicators



Part E - Baseline-versus-Spoof Comparison

Indicator	Baseline	Controlled spoof	Assessment
DNS responder IP	10.230.178.27	10.230.178.27	Spoofed DNS response frame forged the IP source header of the legitimate gateway resolver.
DNS responder MAC	ca:7f:b0:45:62:41 (Gateway MAC)	Analyst MAC	Source IP claims resolver identity, but Ethernet MAC matches Analyst adapter.
Transaction ID match	Matches Query ID (0x1a2b)	Matches Query ID (0x1a2b)	Valid transaction ID race condition win by spoofing engine.
Answer IP	10.230.178.27 (Authorized Host)	10.230.178.196 (Analyst IP)	Forged Record: Response points victim traffic to analyst web server hosting warning banner.
TTL	10800	5	Short fixed TTL enforced by Scapy script to evade caching.
Response time	36ms	< 1 ms (0.000676 s)	Forged response created locally on LAN arrived ahead of legitimate upstream reply.
Competing answers	1 Response Frame	2 Response Frames	Two distinct response packets recorded for a single Transaction ID (Forged packet arrived first).

Gateway ARP mapping	10.230.178.27 -> ca:7f:b0:45:62:41	10.230.178.27 -> Analyst MAC	ARP Cache Poisoning: Gratuitous ARP replies associated gateway IP with Analyst MAC.
Subsequent HTTP destination	Legitimate Portal Endpoint	10.230.178.196:80	Victim immediately initiated TCP handshake and HTTP GET to local analyst web server.

Part F - Cleanup and Verification

